

Healthletics AI Capstone

Step 8: Deployment Guide

Purpose: This guide explains how the Healthletics member-churn prediction model could be prepared for practical deployment while maintaining reproducibility, responsible AI practices, human oversight, and appropriate validation.

1. Deployment Objective

The proposed deployment would use the churn prediction model to identify members who may be at increased risk of discontinuing their membership. The purpose is to support proactive retention activities such as member check-ins, coaching, engagement support, or personalised retention offers. The model should support staff decision-making rather than automatically determine actions for members.

2. Recommended Deployment Workflow

- **1. Collect data:** Use appropriate and authorised member, attendance, and membership information.
- **2. Prepare data:** Apply the same validated preprocessing and feature-engineering workflow used during modelling.
- **3. Generate predictions:** Use the trained model to estimate churn risk for eligible members.
- **4. Review risk:** Present risk information to authorised staff together with relevant member context.
- **5. Engage members:** Use positive, supportive retention actions rather than automatic penalties or restrictions.
- **6. Measure results:** Track retention outcomes and intervention results.
- **7. Monitor the model:** Regularly review performance, fairness, data quality, and changes in member behaviour.

3. Model Deployment Considerations

- **Reproducibility:** Keep preprocessing steps, feature engineering, model configuration, and software dependencies documented.
- **Data consistency:** New production data should use the same feature definitions and formats as the data used during model development.
- **Validation before production:** The public gym dataset should be treated as a proof of concept. The model should be validated and, where appropriate, retrained using representative Healthletics data before operational use.
- **Model versioning:** Record the model version, training data period, features, evaluation results, and deployment date.
- **Fallback process:** Maintain a manual process if the model, data pipeline, or prediction service becomes unavailable.

4. Responsible AI and Human Oversight

The churn score should be treated as an indication of potential risk rather than a definitive prediction of future behaviour. Staff should consider relevant member context before taking action. High-risk predictions should lead to supportive engagement rather than automatic negative decisions. Human accountability should remain part of the process.

5. Privacy and Data Protection

- Only collect and use information necessary for the retention objective.
- Restrict access to authorised staff and systems.
- Protect stored and transferred member data using appropriate security controls.
- Avoid exposing individual member predictions to unauthorised users.
- Define appropriate retention and deletion procedures for model-related data.

6. Fairness Monitoring

After deployment, model performance should be monitored across relevant member groups. Useful measures include recall, precision, false-positive rates, false-negative rates, and prediction rates. Differences should be investigated in context rather than assumed to represent discrimination. Small groups should be interpreted cautiously because performance estimates may be unstable.

7. Model Monitoring

- **Performance:** Accuracy, precision, recall, F1-score and ROC-AUC over time.
- **Data quality:** Missing values, unexpected ranges, new categories and changes in feature distributions.
- **Data drift:** Changes in member behaviour or population characteristics compared with training data.
- **Fairness:** Performance and prediction differences across relevant groups.
- **Business outcome:** Retention rate, churn rate, intervention response and retention uplift.

8. Recommended Pilot Before Full Deployment

- **Phase 1 – Validate:** Test the workflow using representative, appropriately governed Healthletics data.
- **Phase 2 – Pilot:** Use predictions with a limited group of authorised staff and members.
- **Phase 3 – Measure:** Compare retention outcomes and intervention results against an appropriate baseline.
- **Phase 4 – Review:** Assess performance, fairness, privacy, staff workload, and business value.
- **Phase 5 – Scale:** Expand deployment only if the model provides reliable and responsible business value.

9. Business Success Measures

Deployment should ultimately be evaluated using measurable business outcomes rather than model accuracy alone. Potential measures include reduction in member churn, improvement in retention rate, retention uplift among contacted members, intervention response rate, and staff effort required to operate the process. Financial ROI should only be calculated when reliable Healthletics cost and revenue information is available.

10. Deployment Conclusion

The Healthletics churn model provides a potential foundation for a proactive member-retention workflow. However, the current project should be treated as a proof of concept rather than a production-ready system. Before deployment, the model should be validated using relevant Healthletics data, tested for fairness and stability, integrated with appropriate data governance and security controls, and operated with human oversight. A controlled pilot would provide the safest approach for assessing whether the model delivers meaningful retention value in practice.

Key principle: The model should help Healthletics identify where supportive retention attention may be useful—not decide a member's future automatically.